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Remaking Recess Intervention for Improving Peer Interactions at School for Children With Autism Spectrum Disorder: Multisite Randomized Trial

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Abstract. There is a prevailing need for social skills interventions that staff in public schools can deliver effectively to children with autism spectrum disorder (ASD). The present study leveraged partnerships among three large urban school districts and researchers at academic institutions to design and evaluate a social skills intervention, Remaking Recess (RR). In RR, members of the research team coached school personnel on strategies to increase

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peer engagement and social networking during unstructured times (i.e., recess or lunch). A three-site, randomized trial enrolled 80 children with ASD in 69 general education classrooms, grades K–5, in 35 public schools across three large urban districts. Children in RR were more included in peer social networks at follow up than children in the wait-list group based on peer sociometric ratings, $F(1,118) = 1.97$, $p = .05$. While there was no main effect of the intervention on peer joint engagement, children spent less time in solitude during recess in RR than in the wait-list group, $F(1,76) = 4.01$, $p = .049$. School personnel could implement the intervention and found it easy to use in a school setting. These results suggest that a personnel-facilitated intervention holds promise when it comes to changing school social environments and improving social outcomes for children with ASD.

Keywords: community–school collaboration, autism and related disorder, inclusion, intervention, behavior, school engagement

Challenges in social communication and interactions are common and persistent for children with autism spectrum disorder (ASD). These social difficulties are especially challenging at school where children with ASD report feeling lonely and less connected to their peers (Bauminger & Kasari, 2000; Kasari, Locke, Gulsrud, & Rotheram-Fuller, 2011). Despite the need for studies addressing peer interventions in school, most social skills studies are conducted in outpatient clinic settings. The benefits of clinical interventions rarely generalize to the school setting (Bellini, Peters, Benner, & Hopf, 2007). To effect change at school, interventions are likely needed directly within the school setting.

Two approaches to delivering peer social interventions in school settings have been reported. One involves research staff who deliver the intervention on the school's campus. Our research team has engineered two studies in which research staff conducted one-on-one or small-group interventions at school with children with ASD and their neurotypical peers (Kasari et al., 2016; Kasari, Rotheram-Fuller, Locke, & Gulsrud, 2012). Both interventions yielded change on a measure of peer social involvement, one involving peer-reported classroom social networks and the other direct observations of engagement between peers on the playground. The second approach to working in schools is to train the school personnel (teachers and paraprofessionals) to carry out an intervention during the school day to improve peer interactions between children with ASD and their classmates. Most studies that train school personnel to deliver an intervention focus on outcomes of staff fidelity of implementation and improvements in both social initiations and responses of the child with ASD (Brock & Carter, 2013; Koegel, Kim, & Koegel, 2014).

While these studies have demonstrated benefits to children with ASD whether the interventions were delivered by researchers or school personnel, there are a number of important next steps in this line of research. First, in terms of school-based peer interventions, there are very few randomized trials and virtually none that involve school personnel delivering the intervention. To effect long-term and sustainable gains for children with ASD, it will likely be necessary that rigorous group studies are carried out with school personnel delivering the intervention.

Studies find that paraprofessionals in particular have the greatest ability to effect change in peer relationships during unstructured school times (Azad, Locke, Downey, Xie, & Mandell, 2015; Klein Feldman & Matos, 2012; Koegel et al., 2014; Mazurik-Charles & Stefanou, 2010). Paraprofessionals have both close proximity to and responsibility for students on the playground. While studies have shown that some paraprofessional behaviors, like hovering too close, interfering with play, or disengaging can hinder children with ASD from interacting with peers (Giangreco, Edelman, Luiselli, & MacFarland, 1997; Humphrey & Lewis, 2008), other studies report that paraprofessionals can learn to implement effective social interventions for children with ASD with high fidelity (Klein Feldman & Matos, 2012; Koegel et al., 2014; Mazurik-Charles & Stefanou, 2010).

Current Study

The current study comprised a multisite randomized controlled trial involving staff in urban school districts and researchers at academic institutions in three locations (Los Angeles Unified School District, School District of Philadelphia, Rochester City School District) to capture a heterogeneous sample of elementary school–age children in urban and underresourced settings. Remaking Recess (RR), a social skills intervention aimed at improving peer relationships on the playground, was implemented by playground paraprofessionals and supervised by research personnel at each site. We hypothesized that children who received RR would improve their peer engagement as measured by peer social networks and playground observations more than children in a business-as-usual wait-list group. We also explored the extent to which school personnel's fidelity and specific child characteristics (including child age and IQ) would moderate the outcomes.

METHODS

Ninety-seven elementary school–age children were recruited for the study from three large urban school districts (Site 1: Los Angeles Unified School District, Site 2: School District of Philadelphia, Site 3: Rochester City School District). Children were included in the study if they satisfied the following criteria:

1. Child's parents and primary classroom teacher consented to participate,
2. Child had at least one school personnel member who consented to participate on behalf of the child,

3. Child met Autism Diagnostic Observation Schedule research criteria for ASD,
4. Child was between 5 and 11 years old (K–5th grade), and
5. Child was included in a general education classroom for at least 51% of the school day.

Seventeen children did not meet the inclusion criteria, resulting in a total of 80 children in 69 general education classrooms, grades K–5, in 35 public school, as shown in the Consolidated Standards of Reporting Trials diagram (Figure 1). Participants were predominately boys, were racially and ethnically diverse, and had a mean age of approximately 8 years and a mean IQ in the low-average range. Participating school personnel were primarily women, with varying racial and ethnic backgrounds. Detailed child and school personnel characteristics are described in Table 1.

Figure 1. Consolidated Standards of Reporting Trials Chart

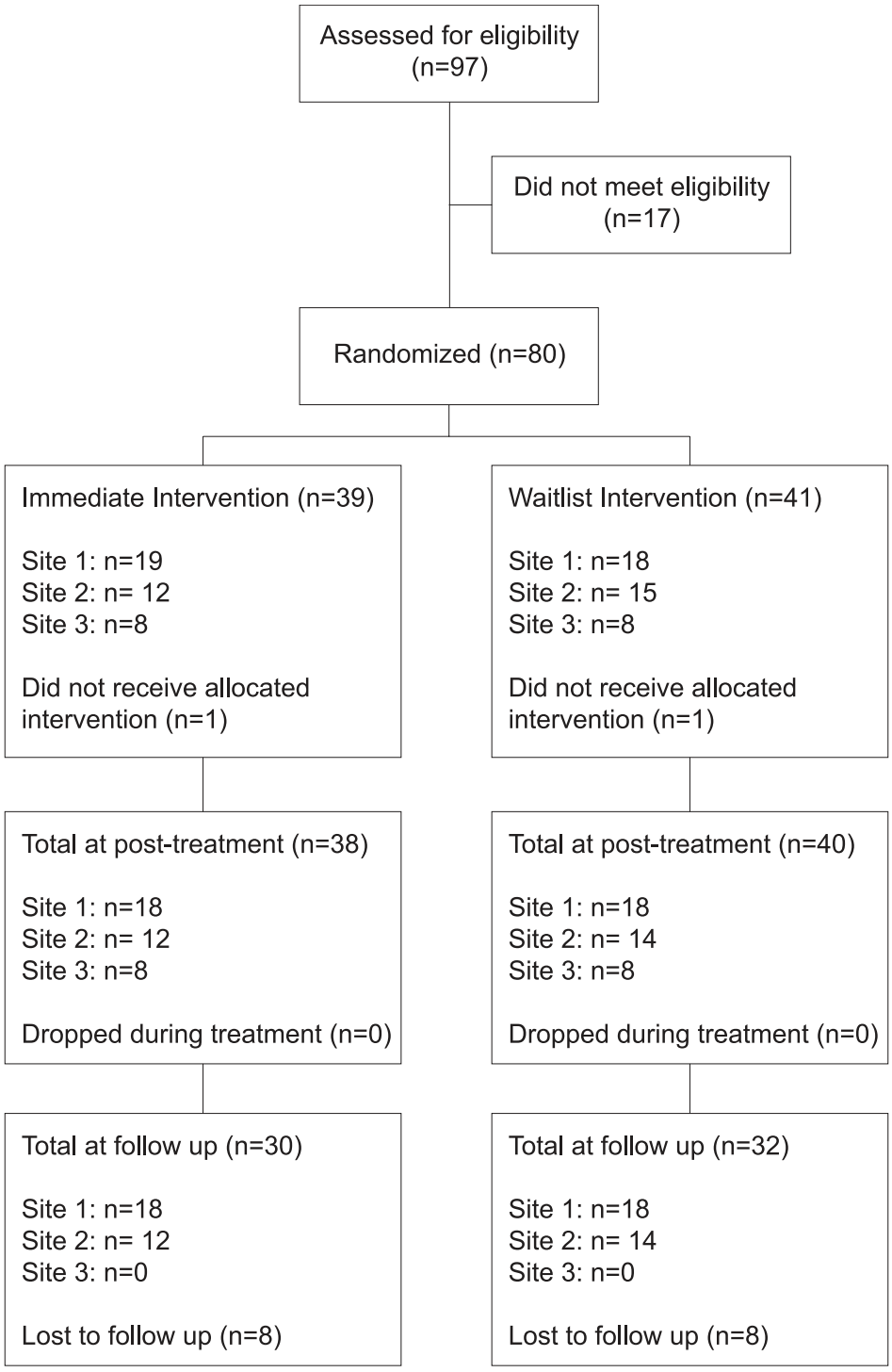


Table 1. Demographic Characteristics: Unadjusted Means and Standard Deviation

Demographic	Remaking Recess ($n = 39$)	Wait List ($n = 41$)	p Value
Child Demographic			
Male: *n* (%)	34 (83)	39 (97.5)	.389
Age in Years (*SD*)	8.31 (1.91)	8.66 (1.59)	.48
Race/Ethnicity: *n* (%)			.11
Caucasian	13 (33.34)	10 (24.39)	
Hispanic	8 (20.51)	3 (7.32)	
African/African American	8 (20.51)	18 (43.90)	
Asian	3 (7.69)	6 (14.63)	
Other/Mixed	4 (10.26)	3 (7.32)	
Do Not Wish to Disclose	3 (7.69)	1 (2.44)	
ADOS Severity (*SD*)	7.7 (1.64)	7.33 (1.8)	.35
DAS			
Verbal IQ (*SD*)	85.67 (17.38)	91.79 (15.32)	.14
Nonverbal IQ (*SD*)	92.08 (18.09)	94.66 (13.23)	.2
IQ (*SD*)	89.74 (18.37)	93.42 (12.56)	.3
Child Outcomes (Preintervention)			
Engagement (%)	37.4 (27.59)	39.6 (29.47)	.73
Solitary (%)	24.5 (24)	19.75 (21.41)	.35
Social Network Salience	33.84 (23.39)	26.98 (24.18)	.23
School Personnel Demographic	Remaking Recess ($n = 40$)	Wait List ($n = 38$)	p Value
Male: *n* (%)	6 (17)	7 (20)	.954
Age in Years (*SD*)	43.38 (15.27)	38.73 (11.49)	.221
Race/Ethnicity: *n* (%)			.798
Caucasian	7 (17.5)	8 (21.05)	
Hispanic	13 (32.5)	7 (18.42)	
African/African American	11 (27.5)	14 (36.84)	
Asian	1 (2.5)	1 (2.63)	
Other/Multiracial	2 (5)	3 (7.90)	
Do Not Wish to Disclose	6 (15)	5 (13.16)	

Note. ADOS = Autism Diagnostic Observation Schedule; DAS = Differential Ability Scales.

Randomization

An independent statistical center conducted the random assignment at the school level within each site, with allocation concealed from all other study personnel. Within each site, schools were matched based on the demographics such as school size, participating grade levels, percentage of students reporting as ethnic minorities, percentage of free and reduced-price lunch, percentage of students with disabilities, percentage of English language learners, and student population in general education classrooms versus special education classrooms. Within the matched sets, random selection distributed participants to the immediate Remaking Recess (RR) treatment group or to a wait-list (WL) control. WL participants received RR during the next academic year. The children, along with 78 school personnel who worked with the children during recess, were randomized to immediate RR treatment ($n = 39$) or WL control ($n = 41$).

Intervention: RR

The RR intervention consisted of 10 sessions of active information sharing and active coaching between school personnel and a coach (a member of the study team) at lunch or recess and 6–10 sessions of reviewing and refining strategies, flexible coaching, and/or trouble shooting. Each session was 20–30 min long. During the session, the coach met with the participating school personnel during recess to help integrate key intervention strategies that support social inclusion and facilitate activities in which the child with ASD could be jointly engaged with peers. The intervention included systematic support fading, in which the coaches worked with the school personnel two to three times a week for the first 3 weeks, two times a week for the fourth and fifth weeks, and one time a week for the following weeks, depending on the school personnel's availability and progress. There were no active coaching sessions during the third month. Table 2 summarizes the content of the coaching sessions.

Coaches included research assistants and doctoral students who had experience working with children with ASD and school personnel. Coaches completed a week-long training in RR that comprised didactics, live demonstrations, and in vivo coaching with performance feedback prior to working with study participants. Two developers of RR examined the fidelity of coaches' implementation (authors JL and MK). All coaches were trained by the treatment developers to a minimum of 80% fidelity using the fidelity checklist. To control for drift, coaches participated in weekly supervision meetings with the intervention developers (authors JL and MK).

Coaches taught school personnel the following skills by providing information, modeling intervention strategies, and providing feedback on implementation during ongoing school activities: (a) identify children who are unengaged on the playground or who have difficulty with peer interaction; (b) use strategies to help children engage with each other (e.g., starting age-appropriate games and activities on the playground); (c) learn when to facilitate and when to fade support to help children begin and maintain interactions with each other; and (d) troubleshoot and refine the intervention. Intervention strategies are detailed in a manual (Kretzmann, Shih, & Kasari, 2015) and illustrated in videos (RemakingRecess.org, 2014). Research coaches who followed the manualized intervention coached school personnel on the playground in the treatment model. Table 2 shows the topics covered in each session.

Assessment

Table 3 summarizes the assessments and their corresponding measures given to children and school personnel from screening to follow up.

Child Measures

Screening/Eligibility Measures: Differential Ability Scales, 2nd Edition

The Differential Ability Scales–Second Edition (DAS-II; Elliot, 2007) is an individually administered battery of cognitive tests for children and adolescents ages 2 years and 6 months through 17 years and 11 months. The DAS-II yields a composite score focused on reasoning and conceptual abilities called the General Conceptual Abilities (GCA) score ($M = 100$, $SD = 15$), which has high internal consistency. The DAS-II data were collected only at entry as a measurement of

Table 2. Remaking Recess Session Topics

Session #	Week #	Session Topics
1	1	Treatment overview, imagining a child's trajectories based on different scenarios, staff communication and cooperation
2	1	Recognizing the value of peer engagement, coding peer engagement
3	1	Priming for peer engagement, transitioning to an engaging activity
4	2	Facilitating social positioning and peer conversations (in transition, at lunch, on playground)
5	2	Facilitating peer conversations, providing games and activities (learning popular developmentally appropriate games for the specific site, meeting child needs)
6	2	Facilitating peer conversations; providing popular, developmentally appropriate games and activities
7	3	Providing games and activities, strategies to boost peer engagement (recruiting players, participating, employing peers to direct and redirect)
8	3	Providing games and activities, strategies to boost peer engagement (fading out, circulating/scanning)
9	3	Staff communication and cooperation, reinforcing good behavior of students and staff
10	4	In vivo social skills instruction, proactively addressing problem behaviors
11–16	4–12	Flex session for review, refinement and troubleshooting

Table 3. Study Time Points and Measurements

Screening Eligibility Assessment	Baseline Preintervention	Exit Postintervention	Follow Up 4–6 Weeks Postintervention
Child • DAS • ADOS	Child • Social network • POPE School Personnel Fidelity • Blinded observer • Coach rated • Self-rated	Child • Social network • POPE School Personnel Fidelity • Blinded observer • Coach rated • Self-rated • Implementation climate	Child • Social network • POPE School Personnel Fidelity • Blinded observer • Coach rated • Self-rated

Note. DAS = Differential Ability Scales; ADOS = Autism Diagnostic Observation Schedule; POPE = Playground Observation of Peer Engagement.

a child's baseline cognitive level (to ensure children had the verbal and nonverbal abilities to fully comprehend components of the intervention).

Primary Outcome: Playground Observation of Peer Engagement

The Playground Observation of Peer Engagement (POPE; Kasari, Rotheram-Fuller, & Locke, 2005) is a time-interval coding system that identifies durations of joint engagement with peers (e.g., having a conversation, playing a game) or solitary play (isolated from peers). Observers who were independent and blinded to treatment group assignment watched the target child on the playground for 40 s consecutively and then coded for 20 s during the recess or lunch play period (an average of 15 min per observation). Playground engagement states were expressed as the percentage of intervals the target child spent in solitary play or jointly engaged with others. Coders were trained to high interrater reliability (>.80) prior to beginning the study by practicing observing children and then using standard videotapes or live peer interactions to test interobserver agreement. Coder drift was assessed and overseen by site coordinators (authors JL and MK and one senior research supervisor CC) at each site (Cohen's kappa = 91, range = .80–.97).

Secondary Outcome: The Friendship Survey

The Friendship Survey is a two-page questionnaire used to assess children's friendships and networks of peer relationships in the classroom (Cairns & Cairns, 1994). Only children with ASD and their typically developing classmates who gave written assent and whose parents gave written consent completed the survey. Surveys were given to the class only after at least half of the class provided consent and assent. Surveys completed by half of the class (reporting on all members of the class) are required to obtain a valid measure of classroom social networks (Cairns & Cairns, 1994).

Children were asked to list the names of their friends. Then they were asked to list all the children in their class who hang out together; this free-recall method was used rather than asking respondents to place each peer's name in a group because the relative strength of students' tendencies to recall classmates is an important indicator of the peers' prominence in the social structure. Following the procedure of Cairns and Cairns (1994), profile scores for each child were derived. Profile scores indicate the number of times that a child was listed as belonging to a group. However, profile scores cannot be directly compared across varying classroom sizes or over time, since the structure of the classroom may also change across time. Hence, the social network salience was created to describe a child's social prominence or salience within his or her own classroom across multiple time points (Kasari et al., 2012) and was calculated for each assessment point by dividing the child's profile score at that assessment by the highest profile score within that child's class at pretreatment. The social network salience controls for classroom differences, such that each participant's score was relative to the scores of their own class within one fixed time point.

Exploratory Outcome: Teacher's Perception of Children's Social Skills and Classroom Conduct

The Teacher Perception of Social Skill and Classroom Conduct is a 26-item questionnaire. Each item is rated on a 3-point Likert scale (1 = *never*, 2 = *sometimes*, and 3 = *very often*). The 12-item social skills domain assesses the child's strengths, such as adaptability to the school classroom and environment, quality of interactions with peers, and popularity or likeability among peers. The 14-item classroom conduct domain assesses the child's challenging behavior, anxiety, and flexibility. The Early Head Start Family and Child Experiences Survey program adapted the measures and reported good internal consistency, ranging from .72 to .88 (Sorongon, Kim, & Zill, 2000), and the measure has been used in another social skills intervention study (Kasari et al., 2012).

School Personnel's Strategy Use

Exploratory Outcome: Remaking Recess Strategy Use

The consistency and quality of RR implementation by school personnel was rated by observers blinded to treatment group assignment. Blind observers were present at the schools

only during the major time points (baseline, exit, and follow up) and only during recess or lunch periods. Coaches and blind observers could not both be present to preserve the "blindness" for the observers.

RR participants were also rated by their coaches and by school personnel themselves (self-report). Each of the seven strategies of RR was scored based on whether or not the strategy was used during the observation (0 = *did not use* and 1 = *used*); if the strategy was used, the observer noted how well school personnel implemented the strategy based on a Likert scale ranging from 1 (*not well*) to 5 (*very well*). The proportion of overall strategy usage was derived by totaling the number of strategies that were observed over the number of possible strategies. Quality of intervention delivery was computed by taking the average quality rating across all intervention strategies that were demonstrated. Prior to data collection, blind observers were trained to 80% agreement with the research coordinators of each site.

Implementation Climate

School personnel in RR rated their perceptions about RR in their schools' implementation climate through a 17-item questionnaire that focused on perceptions of program quality and ease of use (6 questions), school support (8 questions), and global perceptions of climate (3 questions). Each question was rated on a Likert scale from 1 (*not true*) to 5 (*true*). Four questions were reverse scored prior to computation. An overall implementation climate was computed by taking the average of all responses.

Statistical Methods

Pretreatment comparisons between treatment groups were conducted using *t* tests, χ^2 tests, and Fisher's exact test (to assess the success of randomization). Generalized linear mixed models with main effects of treatment (RR and WL) and time (pretreatment, end of treatment, and 1 month follow up), treatment by time interactions, and subject-level random intercepts were used to model the longitudinal trajectories of the outcomes. The models also controlled for child pretreatment IQ and the main effect of site. The models were conducted using SAS MIXED and GLIMMIX procedures for all primary outcomes (i.e., percentage of time spent in joint engagement and percentage of time spent in solitary play) and the secondary outcomes (i.e., social network salience and friendship nominations).

Percentage of time in solitary play showed a bimodal distribution in which one group of children spent little to no time in solitary play (0%–15%) and another group of children spent more time in solitary play (15%–93%). Hence, percentage of time in solitary play was modeled using a hurdle model with random effects to assess the effects of treatment on time spent in solitary play across time points (using the SAS NLMIXED procedure) to account for the two processes (binary, generating the little to no time in solitary play, and truncated Gaussian, generating the large amount of time in solitary play). The hurdle model did not converge with the addition of site effects due to the large number of parameters; therefore, site effects were not included in this analysis.

One site was not able to collect follow-up data due to time constraints within the school calendar year. Hence, effects from pretreatment to posttreatment were evaluated using data from all three sites, and effects from pretreatment to follow up were only evaluated using data from the two sites where follow-up data were collected.

Treatment difference/effect was defined as significant differences in the predicted means between pretreatment and posttreatment or between pretreatment and the 1-month follow up. Site differences were defined as significant differences in treatment effect by site across time. If the site differences were found to be significant, post hoc contrasts were used to determine the effect of each treatment and treatment differences within each site across time. If the site difference was not significant, all site interactions were dropped from the model. Moderators of treatment were evaluated by interacting the potential moderators with treatment difference on child outcomes.

Multiple comparisons for the primary outcomes were not adjusted using the Bonferroni correction because the primary outcomes (i.e., percentage of time spent in joint engagement and percentage of time spent in solitary play) were coded from the same observation and are not independent of each other. Lastly, effect sizes are reported in the Results section by using Cohen's f for evaluating treatment differences and the overall effect of time; effect sizes of 0.10, 0.25, and 0.40 are generally regarded as small, moderate, and large, respectively (Cohen, 1988).

RESULTS

There were no significant differences between treatment groups for any child characteristics or primary outcomes preintervention (Table 2). Descriptive statistics for RR and WL at all time points are shown in Table 4. Site differences were not found to be significant for percentage of time spent in joint engagement/games with rules, $\chi^2(15) = 11.5$, $p = .72$; social network salience, $\chi^2(15) = 12.6$, $p = .63$ and $\chi^2(16) = 11.6$, $p = .77$; received friendship nominations, $F(2,63) = 0.28$, $p = .756$); outward friendship nominations, $F(2,63) = 0.16$, $p = .853$; teachers' rating of the children's social skills, $\chi^2(15) = 5.8$, $p = .98$; or teachers' rating of the children's classroom conduct, $\chi^2(15) = 5.5$, $p = .99$. Consequently, site was removed from the final models. Table 4 displays the unadjusted means and standard deviations for the primary and secondary outcomes.

Primary Outcome: Peer Engagement

Percentage of time spent in solitary play was modeled using the hurdle model (i.e., binary and truncated Gaussian). There was a significant treatment difference in the truncated Gaussian model (treatment by time interaction) from

Table 4. Outcomes: Unadjusted Means and Standard Deviations

	Remaking Recess			Wait List		
Outcome	Pretreatment	Posttreatment	Follow Up	Pretreatment	Posttreatment	Follow Up
Child Outcome						
Joint Engagement	37.4 (27.59)	44.48 (24.48)	45.72 (29.75)	39.6 (29.47)	39.11 (26.98)	40.77 (29.75)
Solitary	24.5 (24.00)	14.53 (16.08)	21.51 (20.93)	19.75 (21.41)	21.61 (27.25)	17.63 (20.3)
Social Network Salience	33.85 (23.39)	37.37 (27.54)	43.22 (33.54)	26.98 (24.18)	27.98 (26.21)	26.74 (27.3)
Received Friendship Nominations	1.4 (1.17)	1.94 (1.56)	1.87 (1.72)	0.92 (0.91)	0.95 (0.97)	1.09 (1.16)
Outward Friendship Nominations	3.34 (2.01)	4.03 (2.70)	3.6 (2.65)	3.26 (3.33)	2.76 (2.22)	3.67 (3.29)
Aide Outcome						
Blinded Observer Strategy Use	0.30 (0.26)	0.52 (0.28)	0.40 (0.29)	0.26 (0.23)	0.34 (0.27)	0.33 (0.24)
Blinded Observer Quality of Strategy Use	2.59 (0.72)	3.46 (0.88)	3.20 (0.90)	2.52 (0.76)	2.82 (0.89)	2.94 (0.68)
Coach-Rated Strategy Use	0.36 (0.25)	0.76 (0.27)	0.62 (0.37)	NA	NA	NA
Self-Rated Strategy Use	0.74 (0.22)	0.9 (0.19)	0.85 (0.25)	NA	NA	NA

NA = not applicable.

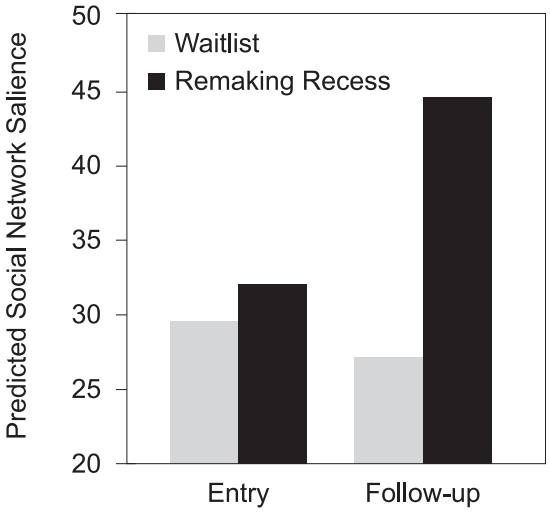
pretreatment to posttreatment, $F(1,76) = 4.01$, $p = .049$, ES = 0.23 (Figure 2). Considering only those children who spent more time in solitary play (i.e., those who crossed the hurdle), the children in the RR group showed significantly more reductions in time spent in solitary play from pretreatment to posttreatment compared to children in the WL group. However, there was no significant treatment difference from pretreatment to 1-month follow up in the truncated Gaussian model, $F(1,76) = 3.14$, $p = .081$, ES = 0.20. In the binary model, there were no significant treatment differences from pretreatment to posttreatment, $F(1,76) = 0.48$, $p = .492$, ES = 0.08, or from pretreatment to 1-month follow up, $F(1,76) = 0.02$, $p = .90$, ES = 0.02.

For the percentage of time spent in joint engagement/games with rules, there were no significant treatment differences (treatment by time interaction), $F(1,71) = 0.96$, $p = .331$, ES = 0.12, and the combined group did not change over time from pretreatment to posttreatment, $F(1,71) = 0.87$, $p = .354$, ES = 0.11 (Figure 3). In addition, there was no

Figure 2. Predicted Mean for Percentage of Time Spent in Solitary Play by Treatment Groups Across Time



Figure 3. Predicted Mean Social Network Salience Scores by Treatment Groups Across Time



significant treatment difference from pretreatment to follow up, $F(1,116) = 0.35$, $p = .555$, ES = 0.05.

Secondary Outcome: Social Network and Social Network Salience

There was no significant difference between treatments or effect of time for the full sample from pre- to posttreatment in children's social network salience, $F(1,67) = 0.30$, $p = .58$, ES = 0.07 and $F(1,67) = 0.45$, $p = .50$, ES = 0.08, respectively. However, there was a significant treatment difference from pretreatment to follow up, where children in RR, on average, had become more salient in their classrooms from pretreatment to follow up than children in the WL group, $F(1,118) = 1.97$, $p = .05$, ES = 0.13 (Figure 3).

Social Network and Friendship Nominations

Children's received friendship nominations did not change across time, $F(1,67) = 1.34$, $p = .251$, ES = 0.14, and did not differ between treatment groups, $F(1,63) = 1.03$, $p = .314$, ES = 0.13. However, there was a significant treatment difference in children's outward friendship nominations between treatment groups, with children in RR having greater outward friendship nominations compared to the children in WL from pre- to posttreatment, $F(1,67) = 4.52$, $p = .037$, ES = 0.26.

Exploratory Outcome: Perceptions of School Personnel

Children's classroom teachers rated the children's social skills and classroom conduct across time. There were no treatment differences in either children's social skills, $F(1,66) = 0.89$, $p = .35$, ES = 0.12, or classroom conduct, $F(1,64) = 0.33$, $p = .567$, ES = 0.07. However, on average, teachers' perceptions of the children's social skills improved from pretreatment to posttreatment [$F(1,66) = 10.81$, $p < .001$, ES = 0.40], and the teachers' perceptions of the children's classroom conduct, remained stable for both treatment groups, [$F(1,64) = 1.20$, $p = .278$, ES = 0.14].

Exploratory Moderators of Treatment on Children's Outcomes

Several potential moderators of treatment were examined. It was found that children's IQ, age, and grade level did not moderate the effect of treatment on outcomes.

Exploratory Outcomes: School Personnel's Strategy (Blinded Observers, Coaches, and Self-Report) and Implementation Climate

Blinded Observers

On average, school personnel from both treatment groups showed significant increases in strategy use, $F(1,56) = 15.74$, $p < .001$, ES = 0.53, but did not significantly differ from each other from pre- to postintervention, $F(1,56) = 2.88$, $p = .095$, ES = 0.23. School personnel in the RR group increased from using 30% of the strategies at preintervention to 52% of the strategies at postintervention, while school personnel in the WL group increased from using 26% of the strategies at preintervention to 34% of the strategies at postintervention. Although there were no treatment differences in strategy use, school personnel in the RR group showed greater increases in implementation quality from pre- to postintervention compared to those in the WL group, $F(1,54) = 5.54$, $p = .022$, ES = 0.32. On average, implementation quality for school personnel in the RR group increased from 2.6 (out of 5) at preintervention to 3.5 at postintervention, and implementation quality for school personnel in the WL group increased from 2.5 (out of 5) at preintervention to 2.8 at postintervention.

Coach Ratings

On average, coaches rated the school personnel as using 36% of the strategies at preintervention, 76% at postintervention, and 62% at the 1-month follow up. The percentage of strategy use did change significantly over time across sites from pre- to postintervention, $F(1,23) = 71.23$, $p < .001$, ES = 1.75, although there was a significant interaction between site and time, $F(2,23) = 9.11$, $p = .0012$; Site 2 had the greatest increase in overall strategy use from pre- to postintervention, followed by Site 3 and then Site 1.

School Personnel Self-Ratings

Similar to coach ratings of school personnel's strategy use, only school personnel who received the RR intervention coaching self-reported strategy use. School personnel in the WL did not provide self-reported strategy use, since they did not receive coaching.

School personnel self-reported using 74% of the strategies at preintervention, 90% at postintervention, and 85% at the 1-month follow up. There was no significant time by site interaction effect, $F(2,28) = 0.61$, $p = .55$, but there was an overall increase in percentage of strategy use from pre- to postintervention across sites, $F(1,28) = 17.88$, $p < .001$, ES = 0.80.

Implementation Climate

The overall implementation score ranged from 1.23 to 4.88 with a mean of 3.89 and a standard deviation of 0.76. Personnel scored the highest on having a clear understanding of the reason for implementation ($M = 4.69$ out of 5) and the lowest on receiving support on the playground intervention from the classroom team and supervisors ($M = 2.31$ out of 5).

DISCUSSION

The goal of this multisite study was to determine if nonspecialists (school personnel) could implement a peer social skills intervention in vivo and improve the peer relationships of children with ASD. This study yielded three main findings.

First, our primary aim was to increase the amount of time children with ASD spent engaged with their peers on the playground at school. The intervention was mediated through staff tasked with supervising children on the playground, namely the paraprofessional aides. The 8-week intervention resulted in mixed findings on improving peer engagement. Children in the RR intervention did not significantly differ from the control group on peer engagement on the playground at the end of intervention, but the amount of time they were observed in solitary play did decrease. Thus, children in RR were likely increasing in engagement (moving from solitary to parallel or parallel-aware status) but were not fully engaged in joint games and conversations. These findings are consistent with our earlier peer-mediated intervention (Kasari et al., 2012) but inconsistent with the improvements in peer engagements observed in a pilot study of RR (Kretzmann et al., 2015). The dependable decrease in solitary play across studies may indicate that this engagement state is the most sensitive to change, whereas the varying results for peer engagement may indicate that it is more challenging to make improvements in the engagement state. The varying results may also reflect the heterogeneity in contextual factors affecting outcomes discussed herein (Kretzmann et al., 2015). Longer interventions, increased intensity, and added social skill modules may be needed to improve joint engagement between peers.

A second finding was that the children in the RR intervention improved in their connections in the classroom as reported by classmates. Thus, they moved from isolation or being peripheral in social groups to being more prominent in those groups. However, while children in RR nominated more friends than those in the WL group, the friendships were not always reciprocated by their classmates, as received friendship nominations from classmates for children in the RR group remained low and stable. These findings are contrary to our earlier study on peer-mediated intervention (Kasari et al., 2012) in which children with ASD did not nominate more children as friends, but their classmates were more likely to nominate them as a friend at the end of intervention. Thus, the picture of social inclusion of children with ASD remains a complicated one that is likely influenced by multiple factors, including school context, staff and child characteristics, implementation fidelity, and implementation climate. It is also likely that stronger friendship development may be fostered through more direct instruction, either in the school context or in homes with helpful parents who set up successful play dates (Frankel et al., 2010).

Our third main finding concerned the extent to which we successfully taught paraprofessionals to carry out the RR intervention in their school contexts. Here coaches' ratings of fidelity and paraprofessionals' own accounts differed. Paraprofessionals reported using the RR strategies three quarters of the time before intervention began, while coaches rated their use of strategies on the playground at one third of the time. Coaches found that by the end of the intervention, paraprofessionals were indeed using the strategies about three quarters of the time, but this is less than the 90% reported by the paraprofessionals themselves. Despite these obvious differences, paraprofessionals rated the intervention strategies as easy to implement but challenging in a school context that did not support their use of the intervention strategies.

School climate and implementation fidelity may be heavily influenced by policy challenges experienced by urban schools (Locke et al., 2015). For example, all three urban school districts in the current study were experiencing fiscal challenges when the intervention was in session. Staff reductions often led to high adult-to-child ratios, as high as 1 adult to 150–200 children in some cases (Dean, Iadarola, Huynh, Harwood, & Kasari, in submission), which can affect successful intervention implementation (Massey, Armstrong, Boroughs, Henson, & McCash, 2005). The Philadelphia site eliminated recess for a period of time, and both east coast sites had numerous snow days that prevented outdoor play. While recess might take place in the cafeteria during inclement weather, school rules at times worked against developing peer relationships (e.g., not being able to talk during lunch and recess, not being able to choose where to sit). Other policy and rule barriers included not allowing children to run on the playground or to play at all if they forgot their homework or their parent did not sign a form (e.g., benched during recess). When school personnel were able to implement the intervention, the quality of the implementation was significantly better for the school personnel in RR compared to WL. A future step would be to incorporate strategies into the intervention that allow for improving peer relationships in nontraditional contexts, such as indoor activities and structured environments when interactive behaviors are more restricted.

Limitations of the study include difficulty in obtaining follow-up data in the context of rolling school enrollments. Longer follow ups would provide more information on whether the changes were sustainable and whether children who received RR continued to become more connected to their peers over time. Different contexts at each site influenced the implementation of the intervention. For example, the staffing structure at each site was different; some used one-to-one aides for children with ASD, and others had one aide for multiple children. These site differences were not significantly associated with outcomes, suggesting that staff were able to work around their particular issues in delivering the intervention. In addition, the Friendship Survey cannot capture friendships from other classrooms. Hence, the number of friends may be underestimated. Despite these limitations, the data also hold promise for transforming children's learning environments and ultimately improving child outcomes.

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